Q1. Write down the different types of distributions and their corresponding PDF / PMF function.

1. Discreate Probability Distribution (PMF)
↳ Discrete distributions deal with countable values and are described using a Probability Mass Function (PMF).

(a) Bernoulli Distribution
↳ used for multiple independent Bernoulli Trails

PMF: $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

where n = no. of Trails
p = probability of success

b) Bernoulli Distribution
↳ used for single trial with two outcomes (Success/Failure).

PMF: $P(X=x) = p^x (1-p)^{1-x}$, $x = 0, 1$
where p = probability of success.

(c) Poisson Distribution
↳ used to model number of events in a fixed interval

PMF: $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$

where λ = average rate of occurrence.

2. Continuous Probability Distribution (PDF)
↳ Continuous distributions deal with continuous values and are described using a Probability Density Function (PDF).

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(a) Uniform Distribution.

↳ All values in an interval are equally likely.

PDF: $f(x) = \frac{1}{b-a}$, $a \leq x \leq b$

(b) Normal Distribution

↳ Also called Gaussian distribution

PDF: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

where μ = mean

σ = standard deviation

(c) Exponential Distribution

↳ Used for modelling waiting time between events.

PDF: $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$

where λ = rate parameter.

Q2. What is Random Variable, its types and expectations.

→ Random variable is a function that assigns a numerical value to each outcomes of a random experiment.

It is usually denoted by capital letters like X, Y, Z

eg → If we toss a coin, we can define

$X = 1$ for Head

$X = 0$ for Tail

So, the random variable converts outcomes into numbers.

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Types of Random Variable

(a) Discrete Random Variables
↳ A Random Variable that take countable values.
eg → No. of head in 3 Coin tosses

(b) Continuous Random variables
↳ A Random variable that take uncountable values.
eg → Height of Students.

Expectations (Mean) of Random Variable
↳ The expectation or expected value of a random variable is the long-run average value of the variable.
↳ It is denoted by $E(x)$

(a) For Discreate Random Variable

$$\boxed{E(x) = \sum x P(X = x)}$$

It is calculated by multiplying each value of x with its probability and summing them.

(b) For Continuous Random Variable

$$\boxed{E(x) = \int_{-\infty}^{\infty} x f(x)\, dx}$$

where $f(x)$ is the probability density function

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Q3. Prove sufficiency for the following using factorization.

① Bernoulli Distribution

Let $X_1, X_2, \ldots X_n \rightarrow$ Bernoulli(p)

Step 1: PMF of one observation

$$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}, \quad x_i = 0, 1$$

Step 2: Joint PMF

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$

$$L(p) = p^{\Sigma x_i}(1-p)^{n-\Sigma x_i}$$

Step 3: Factorization

Let $T = \sum_{i=1}^{n} X_i$

Then,

$$L(p) = p^{T}(1-p)^{n-T}$$

This depends on data only through T.

Hence, by Factorization Theorem,

$$T = \Sigma X_i$$

is sufficient for p.

② Poisson Distribution

Let $X_1, X_2, \ldots X_n \rightarrow$ Poisson(λ)

Step 1: PMF

$$P(X_i = x_i) = \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}$$

Step 2: Joint PMF

$$L(\lambda) = \prod \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}$$

$$L(\lambda) = e^{-n\lambda}\lambda^{\Sigma x_i}\prod \frac{1}{x_i!}$$

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Step 3 : Factorization

Let $T = \sum x_i$

Then,

$$L(\lambda) = \underbrace{e^{-n\lambda}\lambda^{T}}_{g(T,\lambda)} \cdot \underbrace{\prod \frac{1}{x_i!}}_{h(x)}$$

Hence, $T = \sum x_i$ is sufficient for λ.

③ Uniform Distribution

Let $X_1, \dots X_n \sim U(0, \theta)$

Step 1 : PDF

$$f(x_i) = \frac{1}{\theta}, \quad 0 < x_i < \theta$$

Step 2 : Joint PDF

$$L(\theta) = \frac{1}{\theta^n}, \quad 0 \le x_i < \theta$$

This can be written using indicator function

$$L(\theta) = \frac{1}{\theta^n} I(0 < x_{(n)} < \theta)$$

where $x_{(n)} = \max(X_1, \dots X_n)$

Step 3 : Factorization

$$L(\theta) = \underbrace{\frac{1}{\theta^n} I(x_{(n)} < \theta)}_{g(T,\theta)} \cdot \underbrace{1}_{h(x)}$$

Hence,

$T = X_{(n)} = \max(X_1, \dots, X_n)$ is sufficient for θ

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4. Normal Distribution $N(\mu, \sigma^2)$

Let $X_1, \dots X_n \to N(\mu, \sigma^2)$

Step 1 : PDF

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

Step 2 : Joint PDF

$$L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n e^{-\frac{1}{2\sigma^2}\Sigma(x_i-\mu)^2}$$

Expand :

$$\Sigma(x_i-\mu)^2 = \Sigma x_i^2 - 2\mu\Sigma x_i + n\mu^2$$

Thus likelihood depends on :

$$\Sigma X_i \text{ and } \Sigma X_i^2$$

Step 3 : Factorization

$$L(\mu, \sigma^2) = g(\Sigma X_i, \Sigma X_i^2, \mu, \sigma^2) \cdot h(x)$$

Hence, $T = (\Sigma X_i \; \Sigma X_i^2)$

is sufficient for (μ, σ^2).

If σ^2 is Known, then

$$T = \Sigma X_i$$

is sufficient for μ.

5. Exponential Distribution

Let $X_1, \dots X_n \to$ Exponential (λ)

Step 1 : PDF

$$f(x) = \lambda e^{-\lambda x_i}, \quad x_i > 0$$

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Step 2 : Joint PDF

$$L(\lambda) = \prod \lambda e^{-\lambda x_i}$$

$$L(\lambda) = \lambda^n e^{-\lambda \sum x_i}$$

Step 3 : Factorization

Let $T = \sum X_i$

$$L(\lambda) = \underbrace{\lambda^n e^{-\lambda T}}_{g(T,\lambda)} \cdot \underbrace{1}_{h(x)}$$

Hence, $T = \sum X_i$ is sufficient for λ

Q4. Given the following dataset, perform K-means clustering (K=2). Assume the initial centroids are $C1 = (2,2)$ and $C2 = (6,6)$. Data points : (1,1), (2,3), (5,7), (6,5), (8,8). Assign each point to the nearest centroids and updates the centroids. Show all step.

K-means clustering (K=2)

$C1 = (2,2)$, $C2 = (6,6)$

Data Points : (1,1), (2,3), (5,7), (6,5), (8,8)

We use Euclidean Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 1 : Compute distance from Initial Points

Distance from $C_1(2,2)$ and $C_2(6,6)$

Point	d from C1	d from C2	Cluster
(1,1)	$\sqrt{(1-2)^2+(1-2)^2} = \sqrt{2}$	$\sqrt{(1-6)^2+(1-6)^2} = \sqrt{50}$	C1
(2,3)	$\sqrt{(2-2)^2+(3-2)^2} = 1$	$\sqrt{(2-6)^2+(3-6)^2} = \sqrt{25}$	C1
(5,7)	$\sqrt{(5-2)^2+(7-2)^2} = \sqrt{34}$	$\sqrt{(5-6)^2+(7-6)^2} = \sqrt{2}$	C2
(6,5)	$\sqrt{(6-2)^2+(5-2)^2} = \sqrt{25} = 5$	$\sqrt{(6-6)^2+(5-6)^2} = 1$	C2
(8,8)	$\sqrt{(8-2)^2+(8-2)^2} = \sqrt{72} = 8.48$	$\sqrt{(8-6)^2+(8-6)^2} = \sqrt{8}$	C2

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Step-2: From clusters

cluster 1 : (1,1) (2,3)

cluster 2 : (5,7), (6,5), (8,8)

Step-3 : Update Centroids

New C1 $\quad C1 = \left(\frac{1+2}{2}, \frac{1+3}{2}\right) = (1.5, 2)$

New C2 $\quad C2 = \left(\frac{5+6+8}{3}, \frac{7+5+8}{3}\right) = \left(\frac{19}{3}, \frac{20}{3}\right) = (6.33, 6.67)$

Iteration 2

Step-1 : Calculate distance from C1 (1.5, 2) and C2 (6.33, 6.67)

Point	d from C1	d from C2	Cluster
(1,1)	1.118	7.78	C1
(2,3)	1.118	5.67	C1
(5,7)	6.10	1.37	C2
(6,5)	5.41	1.70	C2
(8,8)	8.85	2.14	C2

No change in cluster assignment

Thus, the final clusters and centroids are

C1 = (1.5, 2)

C2 = (6.33, 6.67)